Predictive Fault Detection and Analysis of Industrial Gas Turbine Modules using Machine Learning Techniques

Radhakrishnan P, Barathan S, Prof.Jitendra Kumar*

Department of Mathematics, School of Advanced Sciences
Vellore Institute of Technology, Vellore, Tamil Nadu, India

Abstract

- We develop a data-driven framework for **predictive fault detection** in industrial gas turbines using supervised machine learning.
- Three models were compared: **Logistic Regression** (linear baseline), **Random Forest** (ensemble), and **SVM** (nonlinear kernel).
- Input telemetry: RPM, Air Pressure, Fuel Flow Rate, Oil Temperature, Vibrations, Torque, Exhaust Gas Temperature, and Power Output.
- Results indicate models reliably classify normal operation but have limited sensitivity to rare fault events due to class imbalance; Random Forest detected a small number of faults while SVM captured nonlinear patterns.
- Key indicators identified: **Vibration**, **Air Pressure**, and **Fuel Flow Rate** — recommended as priority signals for monitoring and further feature engineering.

Introduction

- Industrial gas turbines operate under severe thermal and mechanical stress; early fault detection reduces downtime and maintenance cost.
- Common faults include **compressor fouling**, **turbine-blade degradation**, and **combustion anomalies**, which may manifest subtly in telemetry.
- Traditional model-based diagnostics struggle with system nonlinearity and multimodal sensor data.
- Machine Learning provides a complementary, data-driven means to: detect patterns, rank influential sensors, and support predictive maintenance pipelines.
- Objective of this work: **construct and evaluate** ML models for early fault detection, identify influential features, and propose practical mitigation steps for class imbalance and temporal modeling.

Literature Review

- Diagnostics evolved from purely model-based approaches (physics/Kalman filters) to hybrid and ML-driven systems that improve fault isolation and adaptability.
- Kalman filters, Bayesian estimators and sliding-mode observers remain effective for sensor-fault isolation and residual generation.
- Neural networks, NARX models and deep-learning approaches capture transient and nonlinear behaviors, useful for degradation classification.
- Multisensor fusion, ensemble learning and feature-fusion strategies increase robustness and interpretability of diagnoses.
- Recent trends emphasize: (1) temporal/hybrid models for sequence-aware faults, (2) anomaly-detection or semi-supervised learning to cope with scarce labeled faults, and (3) operational integration with SCADA for real-time predictive maintenance.
- These prior works motivate our focus on CV-based ranking, feature-ranking agreement (Spearman), and exploring ensemble/temporal extensions in future work.

Dataset Description

- Dataset sourced from Kaggle:
  - **Title:** Gas Turbine Engine Fault Detection Dataset
  - **Source:** Kaggle – by Ziya07
  - **License:** CC BY 4.0 (Creative Commons Attribution 4.0 License)
- Features include: RPM, Air Pressure, Fuel Flow rate , Oil Temperature, Vibrations, Torque, Exhaust Gas Temperature and Power Output.
- Target variable: Fault (0 = Normal, 1 = Faulty).

Methodology

1. Data cleaning and normalization.
2. Exploratory Data Analysis (EDA).
3. Model implementation:
    - Logistic Regression (linear baseline)
    - Random Forest (ensemble)
    - Support Vector Machine (nonlinear kernel)
4. Model evaluation metrics:
    - Accuracy, Precision, Recall, F1-score.
5. Feature importance derived from Random Forest and correlation metrics.

Results: Model Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-score
Logistic Regression	67.99	0.00	0.00	0.00
Random Forest	65.83	12.5	1.12	2.06
Support Vector Machine	67.99	0.00	0.00	0.00

Table: Performance Metrics of the ML Models

Interpretation:

- LR and SVM classify normal states correctly but miss rare faults.
- Random Forest detects one fault but introduces false positives.
- Class imbalance limits overall sensitivity.

Confusion Matrix Comparison

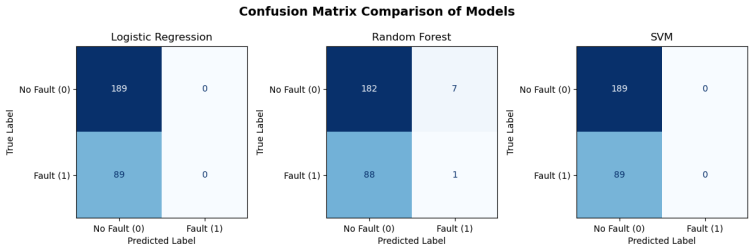


Figure: Confusion Matrices of LR, RF, and SVM Models

Interpretation:

- LR and SVM detect normal operations but miss faults.
- Random Forest identifies one fault but increases false positives.

Feature Importance Analysis

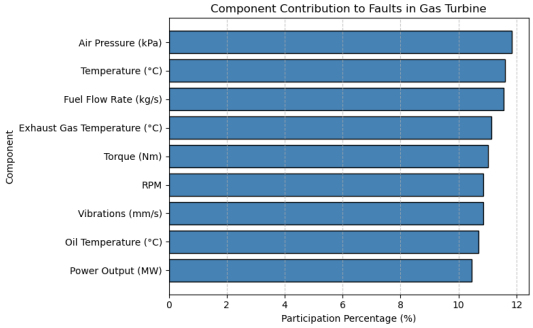


Figure: Component Contribution to Faults in Gas Turbine

Interpretation:

- Air Pressure, Temperature, and Fuel Flow Rate are major contributors.
- Torque, Vibration, and RPM have secondary influence.

Coefficient of Variation (CV) Analysis

Turbine Parameter	CV (%)	CV Rank
Vibrations (mm/s)	24.96	1
Air Pressure (kPa)	12.93	2
Fuel Flow Rate (kg/s)	12.64	3
Power Output (MW)	10.36	4
Oil Temperature (°C)	8.33	5
Torque (Nm)	5.83	6
Exhaust Gas Temperature (°C)	5.77	7
Temperature (°C)	5.48	8
RPM	3.26	9

Table: Coefficient of Variation (CV) and Ranking for Turbine Parameters

Interpretation:

- **Vibrations**, **Air Pressure**, and **Fuel Flow Rate** exhibit the highest variability, showing dynamic operational influence.
- **Temperature** and **RPM** show the lowest CV, indicating stable turbine performance.

Feature Rank Comparison Across Models

Feature	Logistic Rank	Random Forest Rank	SVM Rank
Air Pressure (kPa)	1	1	7
Temperature (°C)	3	2	1
Fuel Flow Rate (kg/s)	9	3	4
Exhaust Gas Temperature (°C)	2	4	6
Torque (Nm)	5	5	3
RPM	4	6	2
Vibrations (mm/s)	7	7	9
Oil Temperature (°C)	8	8	5
Power Output (MW)	6	9	8

Table: Combined Feature Ranking of Gas Turbine Parameters across ML Models

Interpretation:

- Top-ranked parameters: **Air Pressure**, **Temperature**, and **Fuel Flow Rate**.
- Consistency observed between Logistic Regression and Random Forest in key feature ranking.
- SVM exhibits variation due to its nonlinear kernel capturing complex feature dependencies.

Spearman Rank Correlation Analysis

Model Pair	Spearman's Coefficient (ρ)	Correlation Strength
Logistic Regression vs Random Forest	0.55	Moderate
Logistic Regression vs SVM	0.12	Weak
Random Forest vs SVM	0.37	Weak–Moderate

Table: Spearman Rank Correlation among Feature Rankings across ML Models

Interpretation:

- The highest correlation ($\rho = 0.55$) between **Logistic Regression** and **Random Forest** shows moderate agreement, indicating overlapping influential features.
- A low correlation ($\rho = 0.12$) between **Logistic Regression** and **SVM** reflects distinct nonlinear feature behavior in SVM.
- **Random Forest** and **SVM** ($\rho = 0.37$) exhibit partial agreement, supporting the need for hybrid model fusion to enhance reliability.

Findings

- The ML models achieved an overall accuracy between **65–68%**, showing good performance in identifying normal operational states.
- **Random Forest** (65.83%) demonstrated better fault identification ability despite minor accuracy trade-offs.
- **SVM** effectively captured nonlinear relationships but was influenced by data imbalance.
- **Feature importance** revealed that **Air Pressure**, **Temperature**, and **Fuel Flow Rate** are major contributors to turbine performance.
- **Coefficient of Variation (CV)** analysis indicated that **Vibration**, **Air Pressure**, and **Fuel Flow Rate** exhibit the highest variability, representing strong operational sensitivity.
- **Spearman correlation** between Logistic Regression and Random Forest is $\rho = \mathbf{0.55}$, showing a **moderate agreement** in feature relevance.
- A weaker correlation of $\rho = \mathbf{0.12}$ between Logistic Regression and SVM reflects distinct nonlinear dependencies.

Conclusions

- The ML-based framework effectively distinguishes healthy turbine states and provides partial early fault detection.
- **Random Forest** ensures better robustness in identifying fault events, while **SVM** captures nonlinear fault trends.
- Key influencing indicators identified: **Air Pressure**, **Fuel Flow Rate**, and **Vibration**.
- The presence of class imbalance reduced recall and F1-scores, suggesting the need for resampling or cost-sensitive learning.
- **Future Enhancements:**
  - Employ the **Matthews Correlation Coefficient (MCC)** for balanced fault evaluation.
  - Develop **hybrid ensemble and temporal models** to enhance dynamic prediction capability.
  - Integrate the predictive model with **SCADA systems** for real-time monitoring and maintenance scheduling.

References

Marinai, L., Probert, D., & Singh, R. (2004). Prospects for aero gas-turbine diagnostics. *Applied Energy, 79*, 109–126.

Ogaji, S., Li, Y.G., Sampath, S., & Singh, R. (2005). Gas-turbine fault diagnostics: A fuzzy-logic approach. *Applied Energy, 82*, 81–89.

Simani, S. (2005). Fault diagnosis of a simulated industrial gas turbine. *IEEE Trans. Industrial Informatics, 1*(3), 202–216.

Pourbabaee, B., Meskin, N., & Khorasani, K. (2015). Sensor fault detection using hybrid Kalman filter. *IEEE Trans. Control Systems Technology, 24*(4), 1184–1200.

Rahme, S., & Meskin, N. (2015). Adaptive sliding mode observer for sensor fault diagnosis. *Control Engineering Practice, 38*, 57–74.

Cao, Q., Chen, L., & Liu, Z. (2021). Gas turbine degradation classification using deep learning. *International Journal of Energy Research, 45*, 10513–10526.

Chicco, D., & Jurman, G. (2023). MCC should replace ROC-AUC for binary classification. *Biodata Mining, 16*(1), 18–32.

Yan, X., Li, J., & Zhou, H. (2024). Multisensor fusion on hypergraph for fault diagnosis. *IEEE Transactions on Industrial Informatics, 20*, 10008–10018.

Li, X., Zhang, W., & Ding, Q. (2020). Deep learning-based intelligent fault diagnosis methods toward rotating machinery. *Mechanical Systems and Signal Processing, 115*, 213–237.

Data Source:
Ziya07 – Gas Turbine Engine Fault Detection Dataset, Kaggle (CC BY 4.0 License)

Thank You!